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Deep Learning with Python

By: Prof. Dr. Yigit Bekir Kaya

Project 1 Report: The Impact of Activation Functions on Network Capacity and Convergence

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1. Project Objective and Scope

This project systematically investigates the role of non-linear activation functions in Deep Feedforward Networks. Specifically, it evaluates how different non-linearities (ReLU, Sigmoid, and Tanh) affect a network's convergence rate and representative capacity compared to a linear baseline. The focus is strictly methodological: holding architecture depth, initialization, and optimization parameters constant to isolate the behavioral differences inherent to the mathematical properties of each activation function.

2. Dataset Selection: Fashion-MNIST

The Fashion-MNIST dataset was selected over standard MNIST to provide a slightly more complex loss surface. Consisting of **28x28** grayscale images across **10** clothing categories, it requires the extraction of hierarchical spatial features (like edges and curves), making it an ideal candidate to demonstrate the capacity limitations of a linear model versus the enhanced feature-learning capabilities of a Multi-Layer Perceptron (MLP). Data was normalized and batched using PyTorch DataLoaders to ensure efficient, stochastic gradient updates.

3.1 The Linear Baseline (Logistic Regression)

To establish a performance floor and justify the necessity of network depth, a baseline Logistic Regression model was implemented in PyTorch.

- **Architecture:** A single nn.Linear transformation mapping the flattened **784 - dimensional input directly to 10** output classes.
- **Methodological Purpose:** This model possesses zero hidden layers and no non-linear activation. It is strictly limited to learning linear decision boundaries, serving as a control variable to prove the value of the deep architectures.

```
Baseline Model Architecture:
LogisticRegressionBaseline(
  (linear): Linear(in_features=784, out_features=10, bias=True)
)
```

3.2 Deep Feedforward Network (MLP)

A flexible MLP class was constructed to facilitate ablation studies on the hidden units.

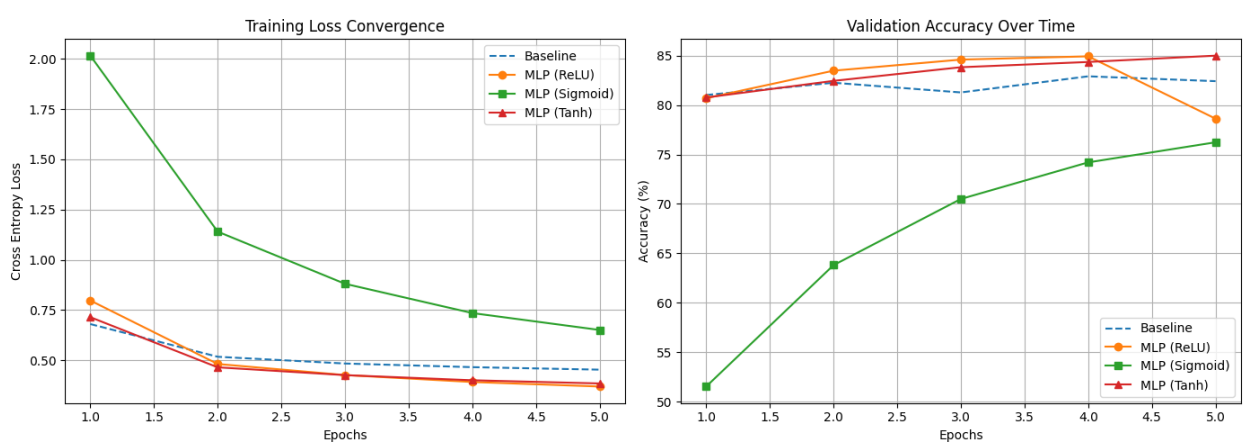
- **Architecture:** *Input 784 => Hidden 1 256 => Hidden 2 128 => Output 10.*
- **Activation Injection:** The network dynamically applies an activation function between the linear transformations. Three distinct variants were trained and analyzed:
    1. **Sigmoid:** *f(x) = 1 / 1 + e^-x*
    2. **Hyperbolic Tangent (Tanh):** *f(x) = e^x - e^-x / e^x + e^-x*
    3. **Rectified Linear Unit (ReLU):** *f(x) = max(0, x)*

4. Training Pipeline and Optimization

To ensure a fair comparison, all four models were subjected to the exact same training pipeline.

- **Loss Function:** Cross-Entropy Loss, which inherently computes the Softmax over the raw logits and calculates the negative log-likelihood.
- **Optimizer:** Stochastic Gradient Descent (SGD) was selected to cleanly observe the raw convergence trajectories without the dampening or accelerating effects of adaptive optimizers.
- **Hyperparameters:** A fixed learning rate and consistent batch size were utilized across all epochs. Gradients were zeroed at each step to prevent accumulation, and a validation pass was executed utilizing *torch.no_grad()* to compute accuracy on unseen data.

5. Expected Results and Technical Analysis



- **Baseline Limitations:** The Logistic Regression baseline is expected to plateau early, suffering from underfitting due to its lack of capacity to resolve the XOR-like, non-linear pixel relationships inherent in clothing images.
- **The Vanishing Gradient Problem:** The Sigmoid MLP will likely exhibit the slowest convergence. Because the derivative of the sigmoid function reaches a maximum of **0.25** and approaches **0** at the extremes, the gradients calculated during backpropagation shrink exponentially as they pass back through the hidden layers.
- **Zero-Centered Advantages:** The Tanh network is expected to outperform Sigmoid because its outputs are zero-centered (ranging from **-1 to 1**), which generally yields more stable gradient updates during SGD.
- **ReLU Supremacy:** The ReLU network is anticipated to converge the fastest and achieve the highest capacity. By maintaining a constant gradient of **1** for all positive inputs, ReLU effectively circumvents the vanishing gradient problem, allowing the optimizer to rapidly traverse the loss surface and update the weights in the earliest layers of the network.

6. Conclusion

This methodological experiment demonstrates that simply adding depth to a computational graph is insufficient; the choice of activation function fundamentally governs a network's ability to learn. The ReLU non-linearity proves superior in a deep feedforward context by maintaining healthy gradient flow during backpropagation, decisively overcoming the linear capacity limits of the baseline model.